

Tiny House

How can we design a tiny house that meets the needs of our clients?

Grade

3-5

Subjects

ELA, Math

Estimated Duration

14-19 instructional
hours

Key Content

area, perimeter,
measurement,
informational
reading,
persuasive/opinion
writing



Student design teams consult with tiny house “clients” to learn about the clients’ budget, needs, and personal preferences. They use this information, along with key mathematics skills related to measurement and geometry, to design and create blueprints of custom tiny homes for their clients. Students also read about housing, tiny homes, and construction and apply what they learn from their reading to their designs. Finally, students use opinion writing and presentations to communicate how their tiny homes meet their clients’ needs. ([View a related project video.](https://www.pblworks.org/video-tiny-house-project)) (<https://www.pblworks.org/video-tiny-house-project>)

Project Path

Project Launch

What goes into creating a dream house?

Entry Event: Create a design for a dream house.

Build Knowledge and Develop & Critique

How can we determine what our clients need to have in a tiny house?

Prepare for and carry out interviews with tiny house clients to learn about their needs.

How are tiny houses planned and built?

Learn key project content, including how homes are built and the key math involved in designing a home.

How can we create tiny houses to meet our clients’ needs?

Create a draft blueprint of a tiny home using information from the client interview.

How can we use writing to communicate how our tiny houses meet our clients’ needs?

Write an opinion paragraph to promote the tiny house design to the client.

Present Products

How can we share our tiny houses with our clients?

Present the tiny house blueprint to the client.

Make It Yours

Reflect on your students

- What kinds of homes do your students live in? How can you connect students' learning about tiny homes to their own experiences with different kinds of buildings?
- How can you use this project to engage students in tactile and kinesthetic learning experiences that help them to understand math?

Reflect on your context

- What types of housing are common in your community? Is there a need for smaller or more economical homes? Would it be possible for students to visit a real tiny house?
- What types of partners (e.g., architects, engineers, building contractors) might you connect students with, virtually or in-person? How can you leverage these partners throughout the project to support knowledge building, critique, and a public audience?
- Whom might you be able to invite in to serve as clients for this project?

Reflect on the content & skills

- How will you break down and scaffold the mathematical skills (i.e., measurement, perimeter, area, operations, etc.) that students will need in order to successfully complete this project?
- How will you help students develop strong interview questions and prepare their presentations for clients?
- What informational texts can you use as resources throughout this project? How can you use these texts to build students' knowledge and deepen their literacy skills?

Take It Further

Here are some ideas for expanding this project if you choose to do so:

- Explore different types of homes in the neighborhood or area surrounding the school with a field trip.
- Create a 3-D model of the tiny home using a range of materials available.
- Invite architects to show how they create a model and/or to share examples of models if available.

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